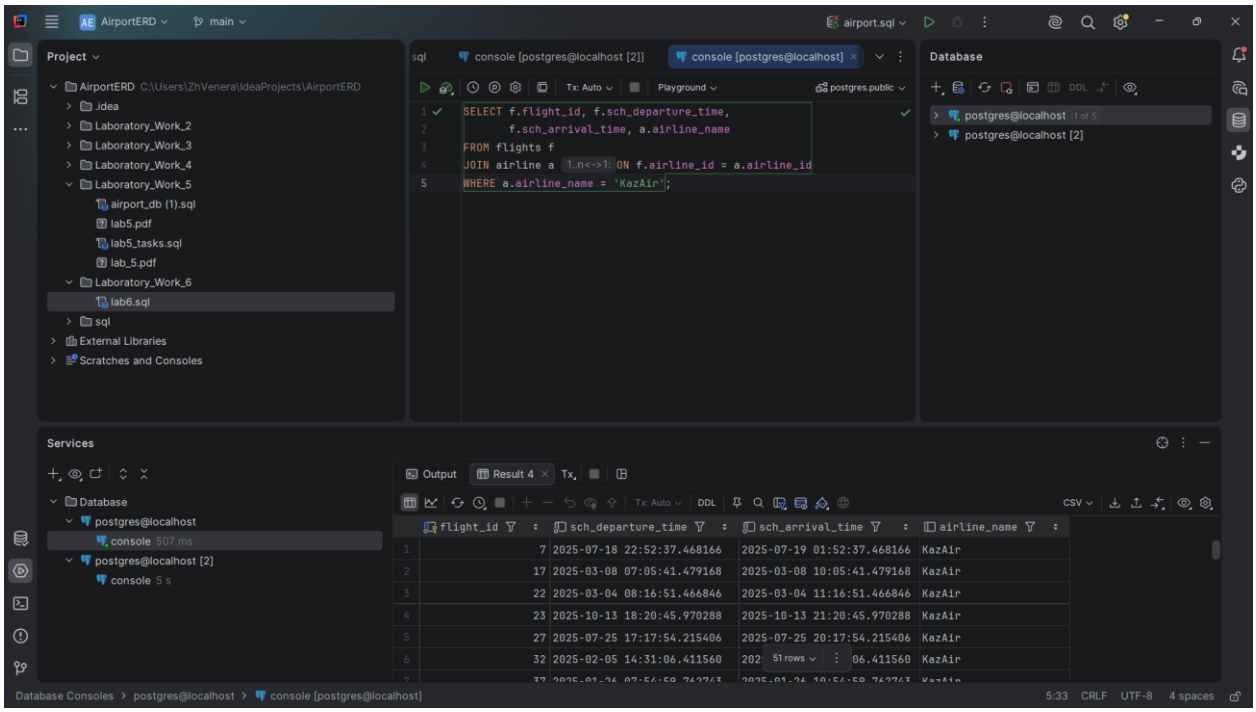
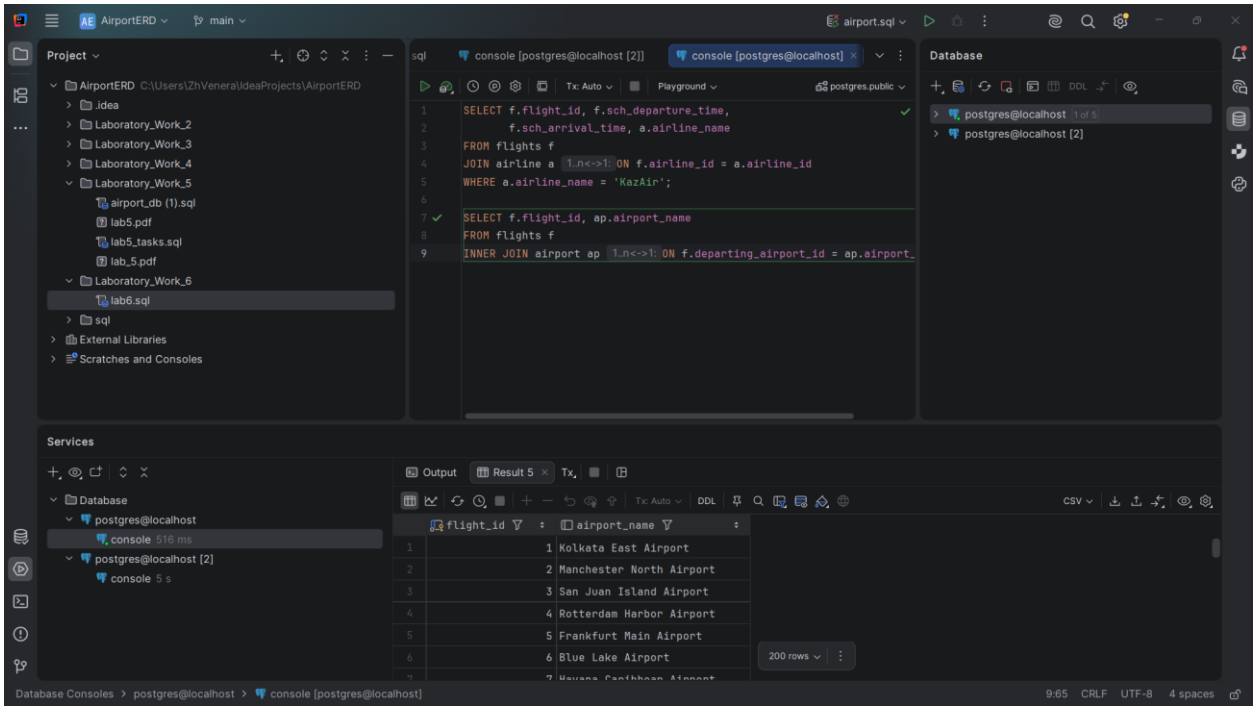
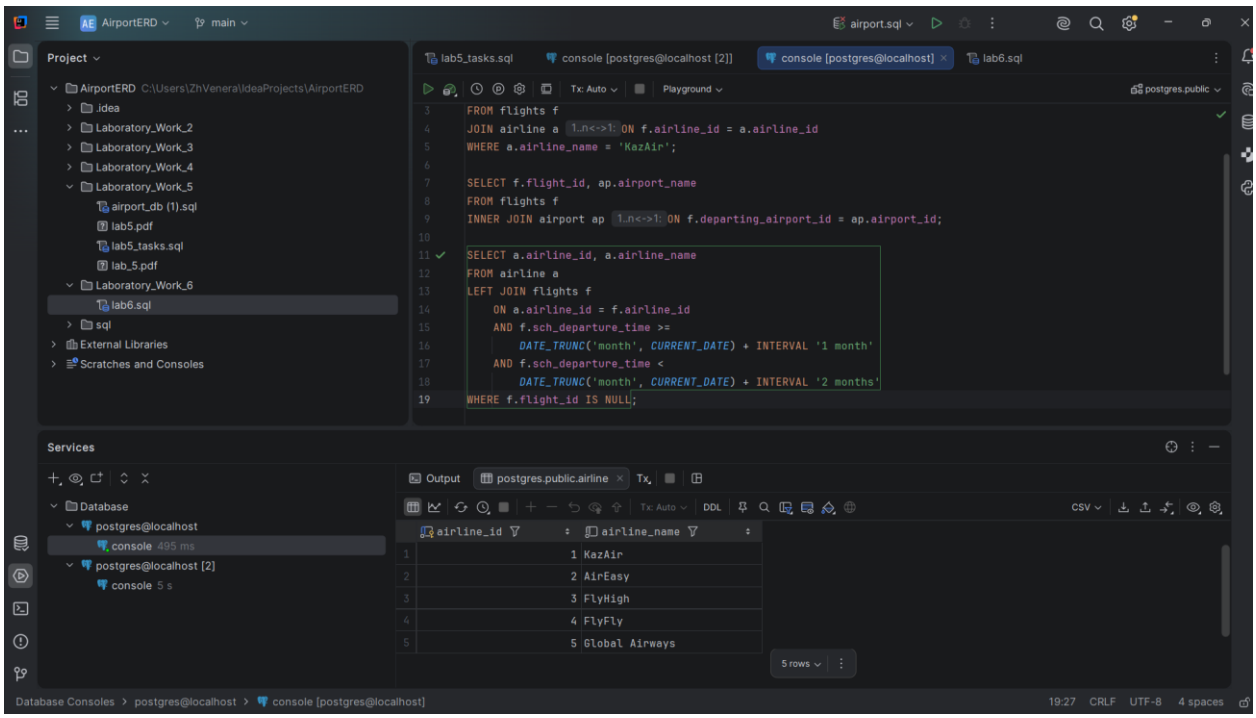
1)


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql
lab5.pdf
lab5_tasks.sql
lab_5.pdf
Laboratory_Work_6
lab6.sql
sql
External Libraries
Scratches and Consoles
console [postgres@localhost [2]]
console [postgres@localhost]
Tx: Auto
Playground
postgres.public
SELECT f.flight_id, f.sch_departure_time,
f.sch_arrival_time, a.airline_name
FROM flights f
JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
WHERE a.airline_name = 'KazAir';
Database
postgres@localhost 1 of 5
postgres@localhost [2]
Services
Output
Result 4
Database
postgres@localhost
console 507 ms
postgres@localhost [2]
console 5 s
DDL
CSV
flight_id
sch_departure_time
sch_arrival_time
airline_name
1 7 2025-07-18 22:52:37.468166 2025-07-19 01:52:37.468166 KazAir
2 17 2025-03-08 07:05:41.479168 2025-03-08 10:05:41.479168 KazAir
3 22 2025-03-04 08:16:51.466846 2025-03-04 11:16:51.466846 KazAir
4 23 2025-10-13 18:20:45.970288 2025-10-13 21:20:45.970288 KazAir
5 27 2025-07-25 17:17:54.215406 2025-07-25 20:17:54.215406 KazAir
6 32 2025-02-05 14:31:06.411560 202 51 rows 06.411560 KazAir
Database Consoles › postgres@localhost › console [postgres@localhost]
5:33 CRLF UTF-8 4 spaces

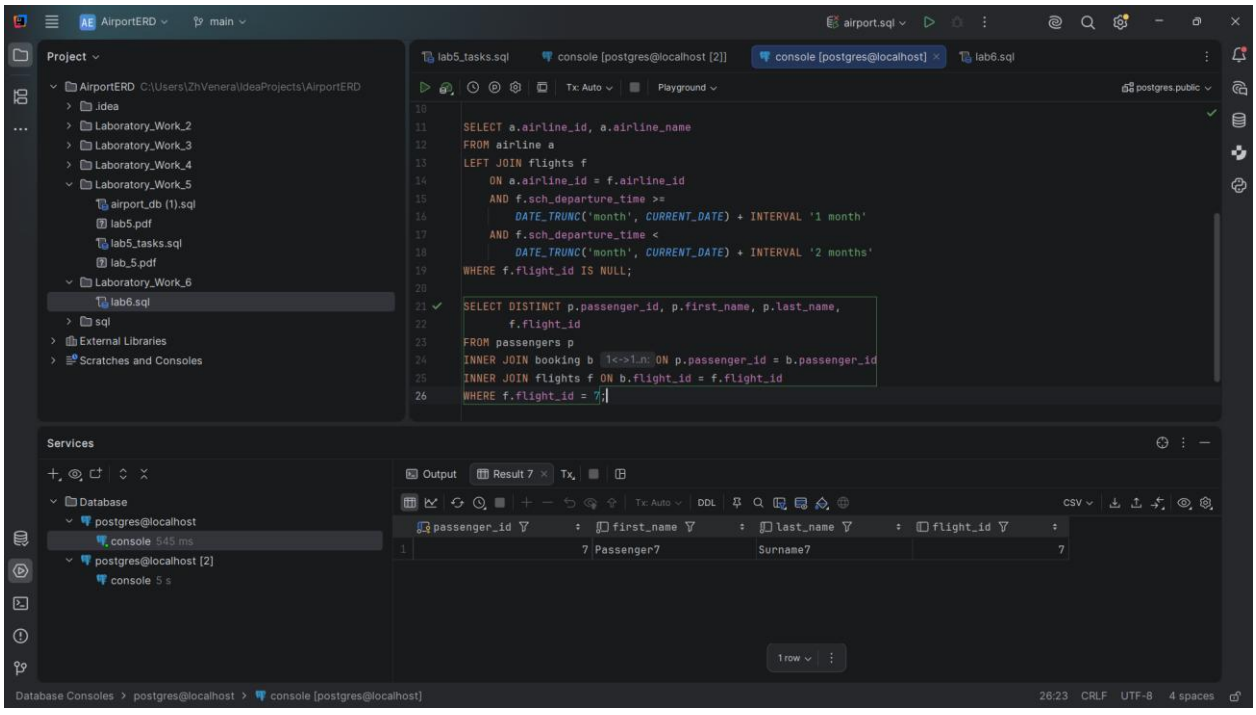
2)


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql
lab5.pdf
lab5_tasks.sql
lab_5.pdf
Laboratory_Work_6
lab6.sql
sql
External Libraries
Scratches and Consoles
console [postgres@localhost [2]]
console [postgres@localhost]
Tx: Auto
Playground
postgres.public
SELECT f.flight_id, f.sch_departure_time,
f.sch_arrival_time, a.airline_name
FROM flights f
JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
WHERE a.airline_name = 'KazAir';
SELECT f.flight_id, ap.airport_name
FROM flights f
INNER JOIN airport ap 1..n<->1: ON f.departing_airport_id = ap.airport_
Database
postgres@localhost 1 of 5
postgres@localhost [2]
Services
Output
Result 5
Database
postgres@localhost
console 516 ms
postgres@localhost [2]
console 5 s
DDL
CSV
flight_id
airport_name
1 1 Kolkata East Airport
2 2 Manchester North Airport
3 3 San Juan Island Airport
4 4 Rotterdam Harbor Airport
5 5 Frankfurt Main Airport
6 6 Blue Lake Airport
200 rows
Database Consoles › postgres@localhost › console [postgres@localhost]
9:65 CRLF UTF-8 4 spaces

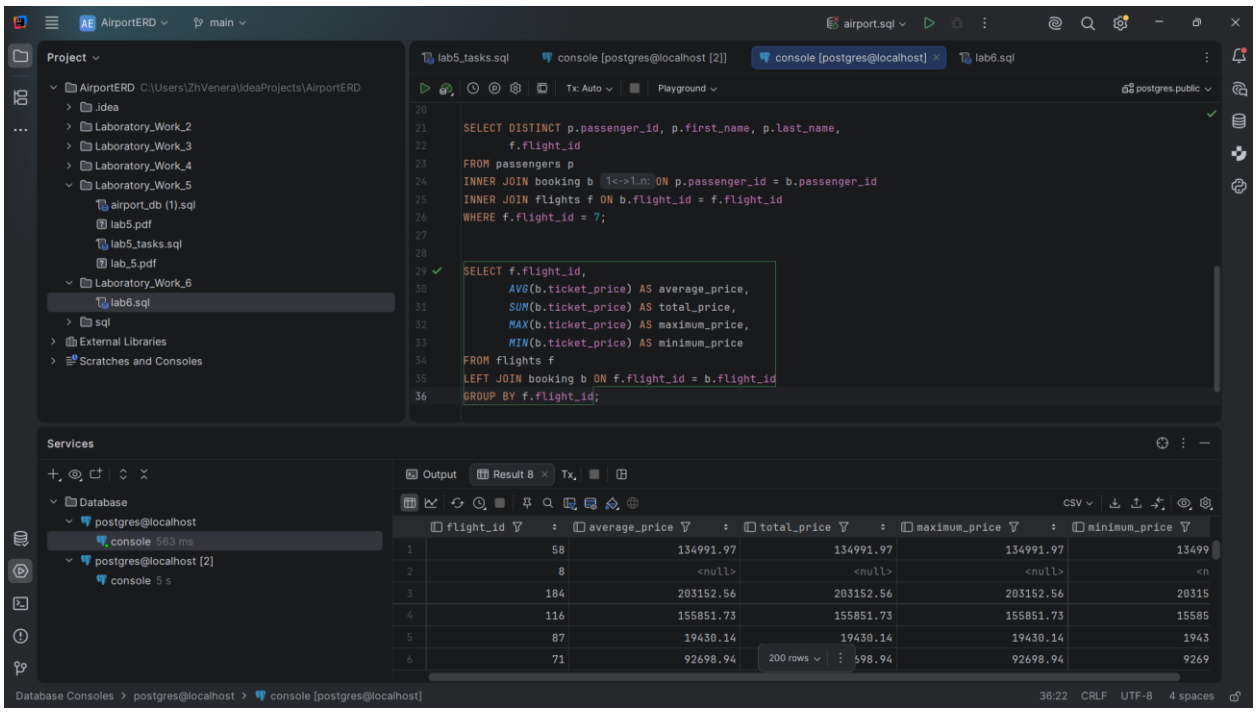
3)


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql
lab5.pdf
lab5_tasks.sql
lab_5.pdf
Laboratory_Work_6
lab6.sql
sql
External Libraries
Scratches and Consoles
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
FROM flights f
JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
WHERE a.airline_name = 'KazAir';
SELECT f.flight_id, ap.airport_name
FROM flights f
INNER JOIN airport ap 1..n<->1: ON f.departing_airport_id = ap.airport_id;
SELECT a.airline_id, a.airline_name
FROM airline a
LEFT JOIN flights f
ON a.airline_id = f.airline_id
AND f.sch_departure_time >=
DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '1 month'
AND f.sch_departure_time <
DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '2 months'
WHERE f.flight_id IS NULL;
Services
Output
postgres.public.airline
Tx
DDL
CSV
Database
postgres@localhost
console 495 ms
postgres@localhost [2]
console 5 s
airline_id
airline_name
1 KazAir
2 AirEasy
3 FlyHigh
4 FlyFly
5 Global Airways
5 rows
Database Consoles
postgres@localhost
console [postgres@localhost]
19:27
CRLF
UTF-8
4 spaces

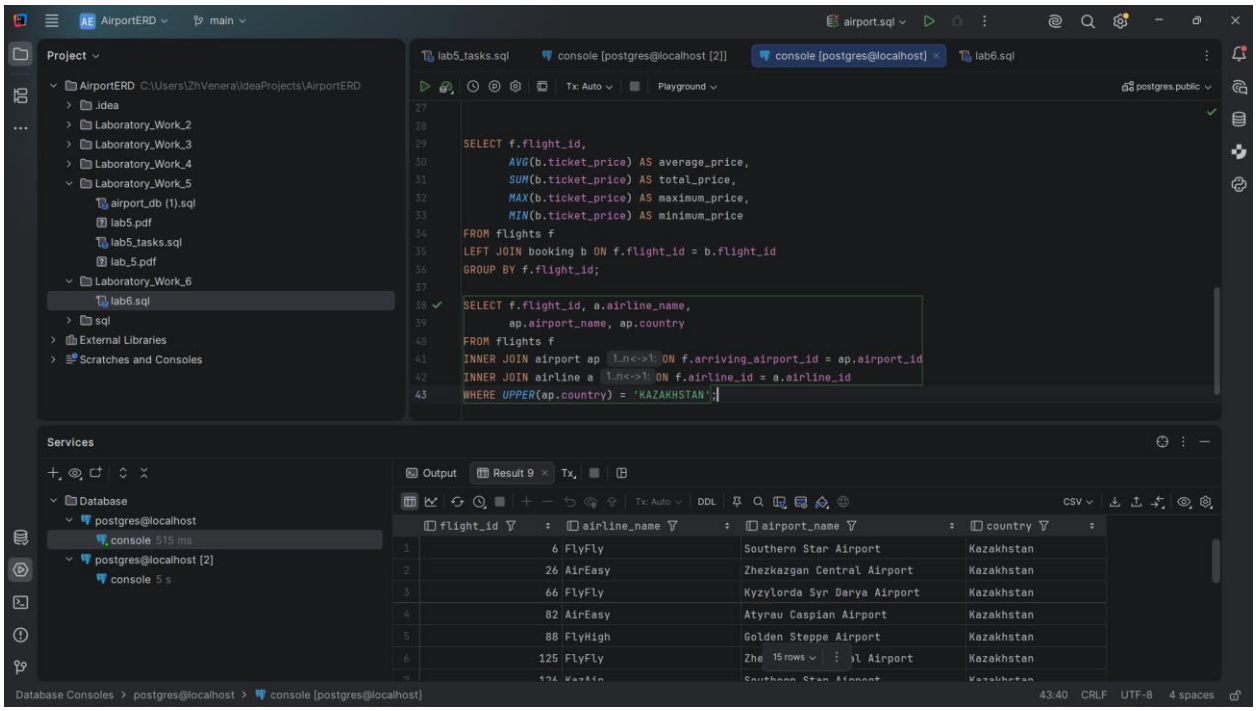
4)


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql
lab5.pdf
lab5_tasks.sql
lab_5.pdf
Laboratory_Work_6
lab6.sql
sql
External Libraries
Scratches and Consoles
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
SELECT a.airline_id, a.airline_name
FROM airline a
LEFT JOIN flights f
ON a.airline_id = f.airline_id
AND f.sch_departure_time >=
DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '1 month'
AND f.sch_departure_time <
DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '2 months'
WHERE f.flight_id IS NULL;
SELECT DISTINCT p.passenger_id, p.first_name, p.last_name,
f.flight_id
FROM passengers p
INNER JOIN booking b 1<->1..n: ON p.passenger_id = b.passenger_id
INNER JOIN flights f ON b.flight_id = f.flight_id
WHERE f.flight_id = 7;
Services
Output
Result 7
Tx
DDL
CSV
Database
postgres@localhost
console 545 ms
postgres@localhost [2]
console 5 s
passenger_id
first_name
last_name
flight_id
7 Passenger7 Surname7 7
1 row
Database Consoles
postgres@localhost
console [postgres@localhost]
26:23
CRLF
UTF-8
4 spaces

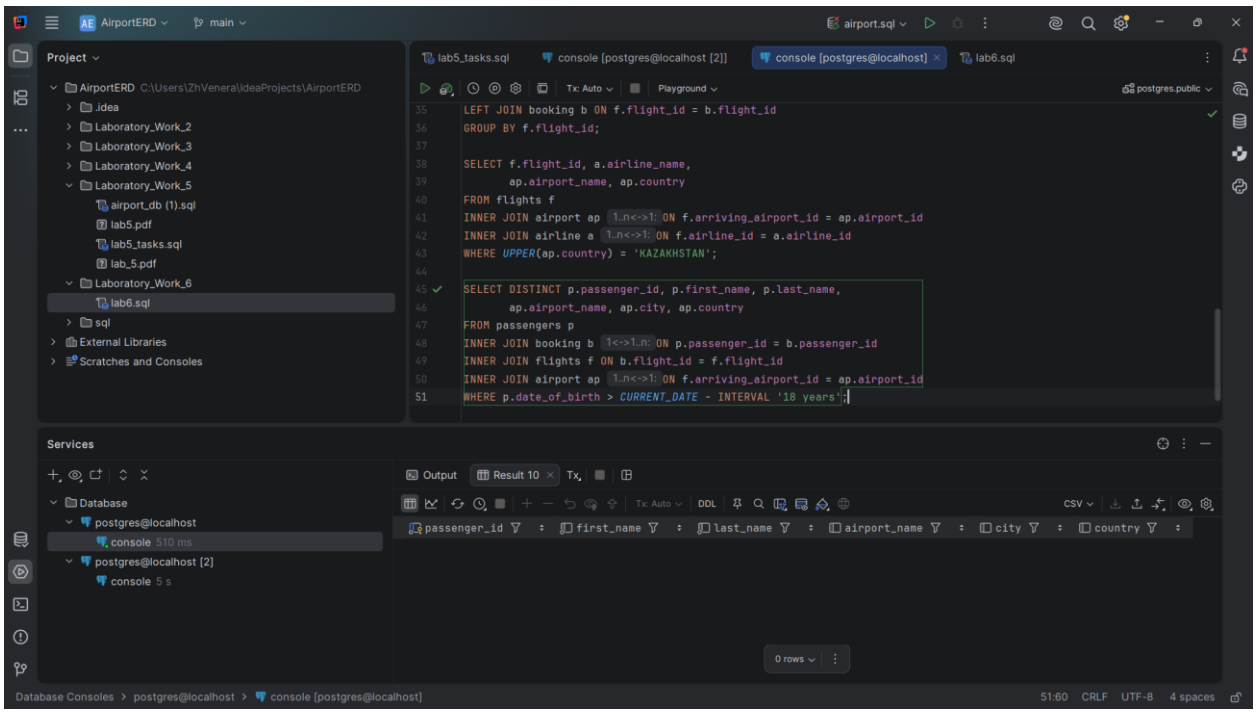
5)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
SELECT DISTINCT p.passenger_id, p.first_name, p.last_name,
f.flight_id
FROM passengers p
INNER JOIN booking b ON p.passenger_id = b.passenger_id
INNER JOIN flights f ON b.flight_id = f.flight_id
WHERE f.flight_id = 7;
SELECT f.flight_id,
AVG(b.ticket_price) AS average_price,
SUM(b.ticket_price) AS total_price,
MAX(b.ticket_price) AS maximum_price,
MIN(b.ticket_price) AS minimum_price
FROM flights f
LEFT JOIN booking b ON f.flight_id = b.flight_id
GROUP BY f.flight_id;
Services
Output
Result 8
flight_id
average_price
total_price
maximum_price
minimum_price
58 134991.97 134991.97 134991.97 13499
8 <null> <null> <null>
184 203152.56 203152.56 203152.56 20315
116 155851.73 155851.73 155851.73 15585
87 19430.14 19430.14 19430.14 1943
71 92698.94 92698.94 92698.94 9269
Database Consoles
postgres@localhost
console [postgres@localhost]
36:22
CRLF
UTF-8
4 spaces

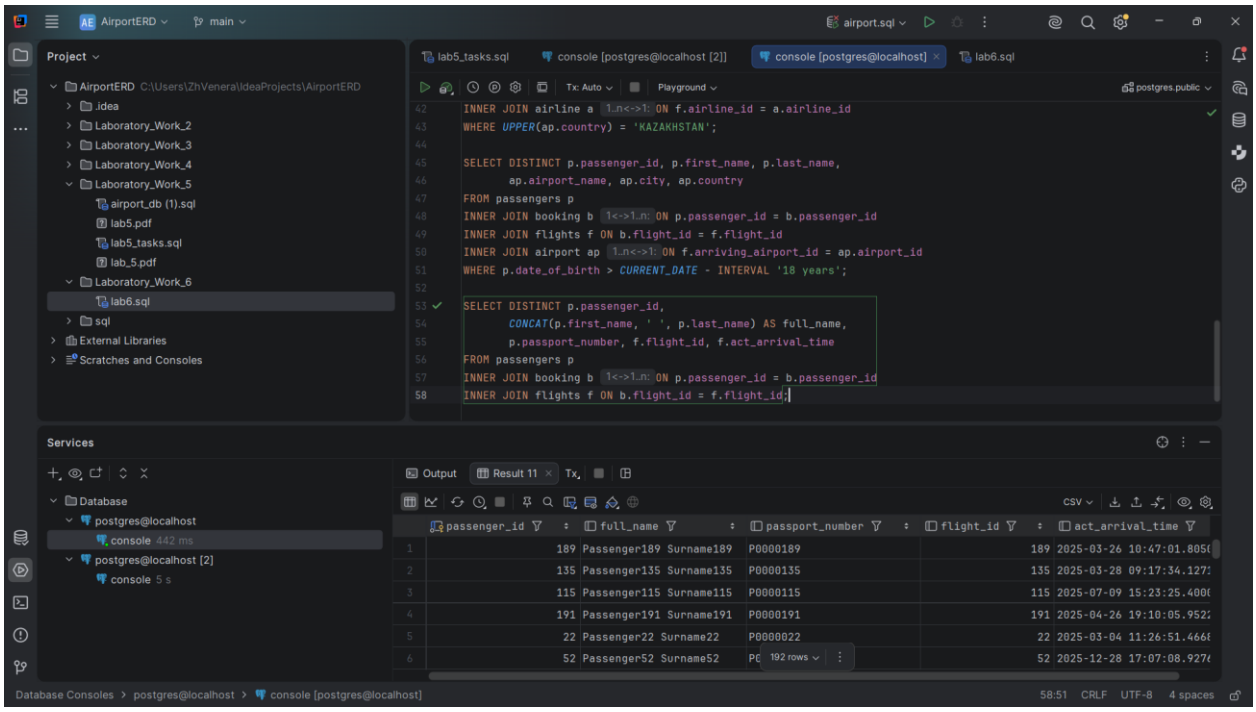
6)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
SELECT f.flight_id,
AVG(b.ticket_price) AS average_price,
SUM(b.ticket_price) AS total_price,
MAX(b.ticket_price) AS maximum_price,
MIN(b.ticket_price) AS minimum_price
FROM flights f
LEFT JOIN booking b ON f.flight_id = b.flight_id
GROUP BY f.flight_id;
SELECT f.flight_id, a.airline_name,
ap.airport_name, ap.country
FROM flights f
INNER JOIN airport ap ON f.arriving_airport_id = ap.airport_id
INNER JOIN airline a ON f.airline_id = a.airline_id
WHERE UPPER(ap.country) = 'KAZAKHSTAN';
Services
Output
Result 9
DDL
flight_id
airline_name
airport_name
country
6 FlyFly Southern Star Airport Kazakhstan
26 AirEasy Zhezkazgan Central Airport Kazakhstan
66 FlyFly Kyzylorda Syr Darya Airport Kazakhstan
82 AirEasy Atyrau Caspian Airport Kazakhstan
88 FlyHigh Golden Steppe Airport Kazakhstan
125 FlyFly Kazakhstan
Database Consoles
postgres@localhost
console [postgres@localhost]
43:40
CRLF
UTF-8
4 spaces

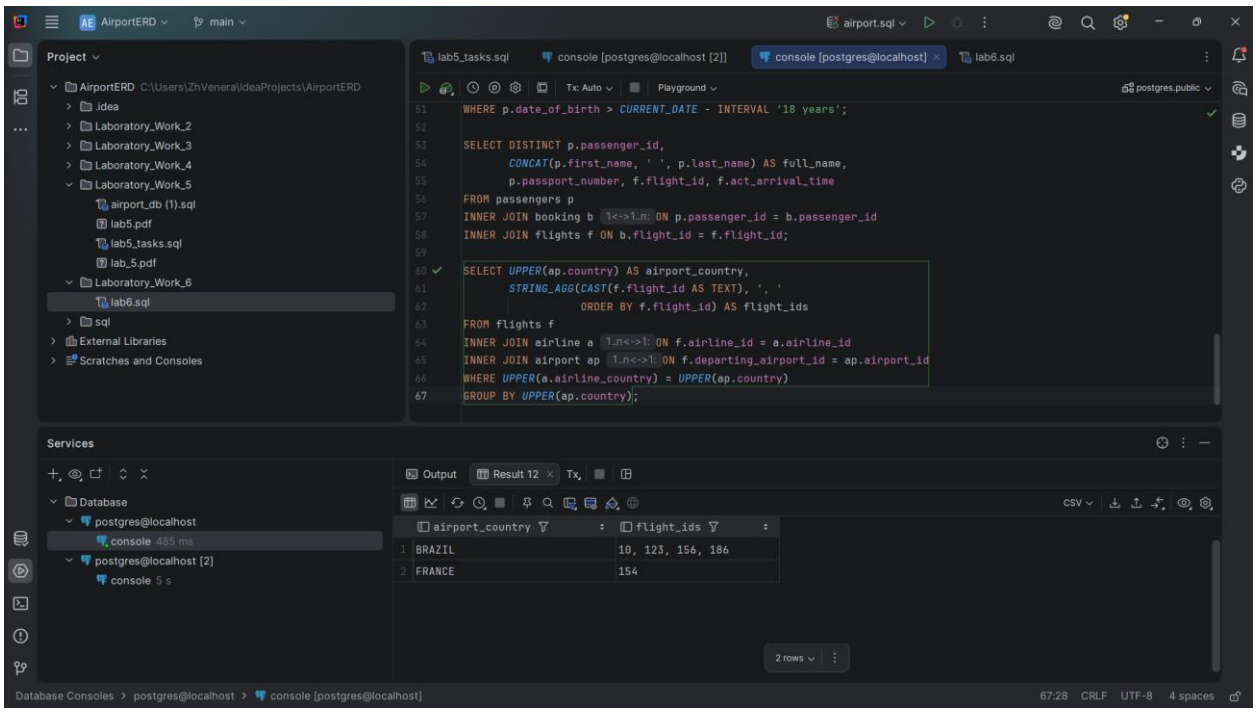
7)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
LEFT JOIN booking b ON f.flight_id = b.flight_id
GROUP BY f.flight_id;
SELECT f.flight_id, a.airline_name,
ap.airport_name, ap.country
FROM flights f
INNER JOIN airport ap 1..n<->1: ON f.arriving_airport_id = ap.airport_id
INNER JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
WHERE UPPER(ap.country) = 'KAZAKHSTAN';
SELECT DISTINCT p.passenger_id, p.first_name, p.last_name,
ap.airport_name, ap.city, ap.country
FROM passengers p
INNER JOIN booking b 1<->1..n: ON p.passenger_id = b.passenger_id
INNER JOIN flights f ON b.flight_id = f.flight_id
INNER JOIN airport ap 1..n<->1: ON f.arriving_airport_id = ap.airport_id
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years';
Services
Output
Result 10
Database
postgres@localhost
console 510 ms
postgres@localhost [2]
console 5 s
passenger_id
first_name
last_name
airport_name
city
country
0 rows
Database Consoles
postgres@localhost
console [postgres@localhost]
51:60
CRLF
UTF-8
4 spaces

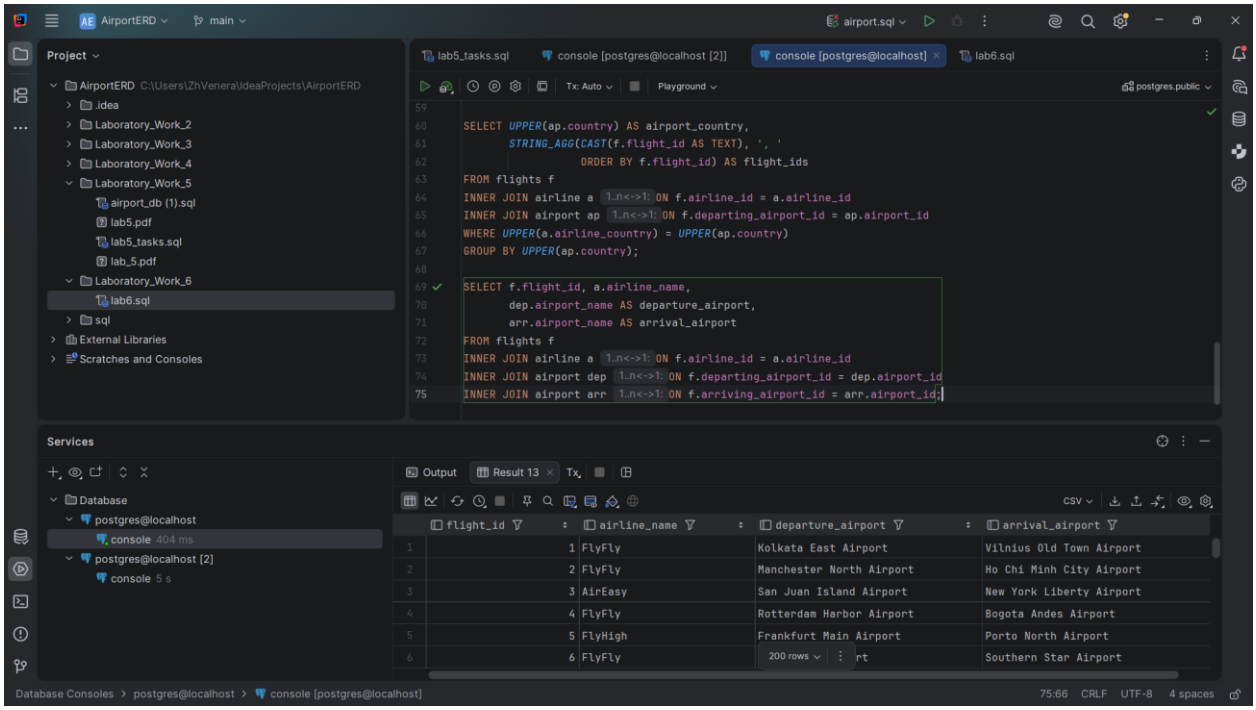
8)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
INNER JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
WHERE UPPER(ap.country) = 'KAZAKHSTAN';
SELECT DISTINCT p.passenger_id, p.first_name, p.last_name,
ap.airport_name, ap.city, ap.country
FROM passengers p
INNER JOIN booking b 1<->1..n: ON p.passenger_id = b.passenger_id
INNER JOIN flights f ON b.flight_id = f.flight_id
INNER JOIN airport ap 1..n<->1: ON f.arriving_airport_id = ap.airport_id
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years';
SELECT DISTINCT p.passenger_id,
CONCAT(p.first_name, ' ', p.last_name) AS full_name,
p.passport_number, f.flight_id, f.act_arrival_time
FROM passengers p
INNER JOIN booking b 1<->1..n: ON p.passenger_id = b.passenger_id
INNER JOIN flights f ON b.flight_id = f.flight_id;
Services
Output
Result 11
Database
postgres@localhost
console 442 ms
postgres@localhost [2]
console 5 s
passenger_id
full_name
passport_number
flight_id
act_arrival_time
189 Passenger189 Surname189 P0000189 189 2025-03-26 10:47:01.805
135 Passenger135 Surname135 P0000135 135 2025-03-28 09:17:34.127
115 Passenger115 Surname115 P0000115 115 2025-07-09 15:23:25.400
191 Passenger191 Surname191 P0000191 191 2025-04-26 19:10:05.952
22 Passenger22 Surname22 P0000022 22 2025-03-04 11:26:51.466
52 Passenger52 Surname52 P0 52 2025-12-28 17:07:08.927
192 rows
Database Consoles
postgres@localhost
console [postgres@localhost]
58:51
CRLF
UTF-8
4 spaces

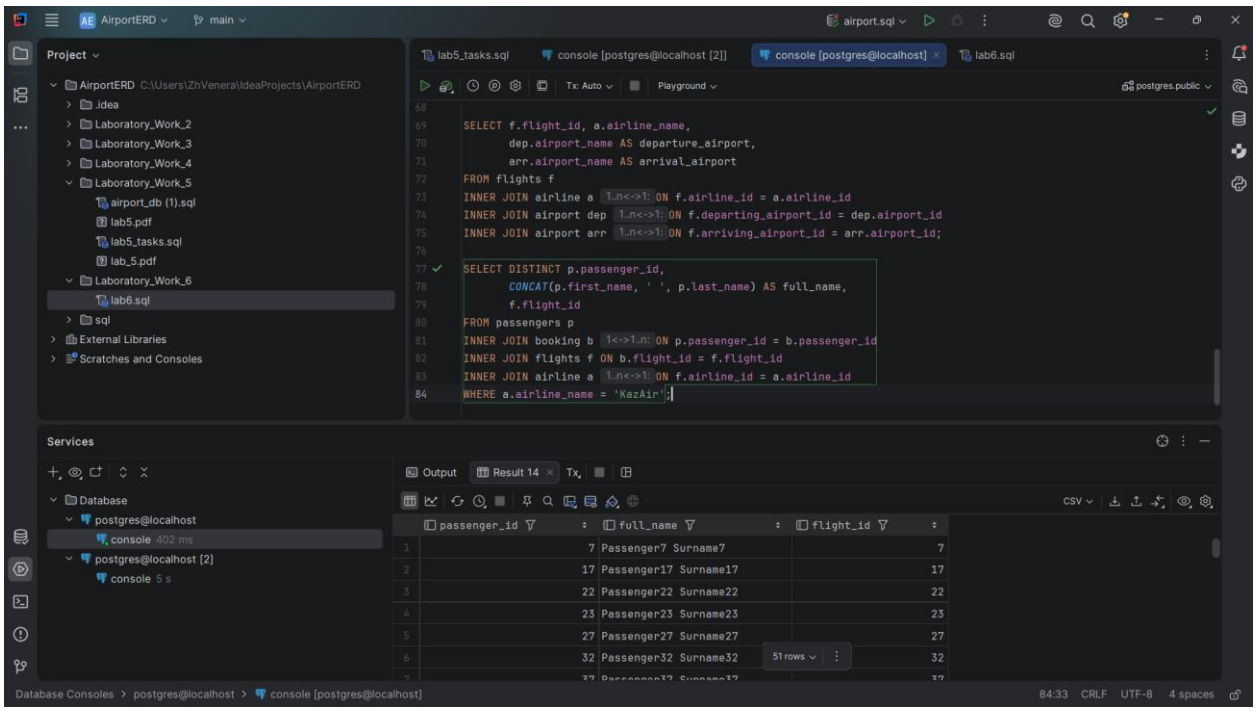
9)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years';
SELECT DISTINCT p.passenger_id,
CONCAT(p.first_name, ' ', p.last_name) AS full_name,
p.passport_number, f.flight_id, f.act_arrival_time
FROM passengers p
INNER JOIN booking b 1<->1..n ON p.passenger_id = b.passenger_id
INNER JOIN flights f ON b.flight_id = f.flight_id;
SELECT UPPER(ap.country) AS airport_country,
STRING_AGG(CAST(f.flight_id AS TEXT), ', '
ORDER BY f.flight_id) AS flight_ids
FROM flights f
INNER JOIN airline a 1..n<->1.. ON f.airline_id = a.airline_id
INNER JOIN airport ap 1..n<->1.. ON f.departing_airport_id = ap.airport_id
WHERE UPPER(a.airline_country) = UPPER(ap.country)
GROUP BY UPPER(ap.country);
Services
Output
Result 12
Tx
CSV
airport_country
flight_ids
1 BRAZIL 10, 123, 156, 186
2 FRANCE 154
2 rows
Database Consoles
postgres@localhost
console [postgres@localhost]
67:28
CRLF
UTF-8
4 spaces

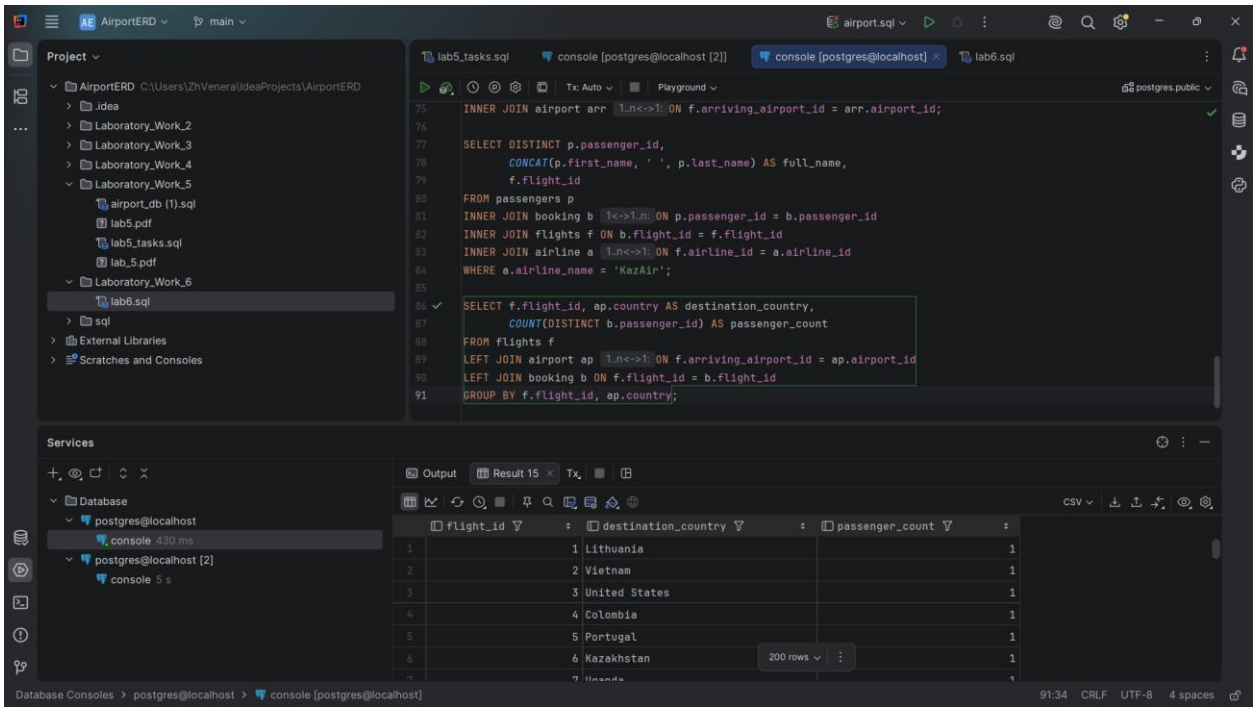
10)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
SELECT UPPER(ap.country) AS airport_country,
STRING_AGG(CAST(f.flight_id AS TEXT), ', '
ORDER BY f.flight_id) AS flight_ids
FROM flights f
INNER JOIN airline a 1..n<->1.. ON f.airline_id = a.airline_id
INNER JOIN airport ap 1..n<->1.. ON f.departing_airport_id = ap.airport_id
WHERE UPPER(a.airline_country) = UPPER(ap.country)
GROUP BY UPPER(ap.country);
SELECT f.flight_id, a.airline_name,
dep.airport_name AS departure_airport,
arr.airport_name AS arrival_airport
FROM flights f
INNER JOIN airline a 1..n<->1.. ON f.airline_id = a.airline_id
INNER JOIN airport dep 1..n<->1.. ON f.departing_airport_id = dep.airport_id
INNER JOIN airport arr 1..n<->1.. ON f.arriving_airport_id = arr.airport_id;
Services
Output
Result 13
Tx
CSV
flight_id
airline_name
departure_airport
arrival_airport
1 1 FlyFly Kolkata East Airport Vilnius Old Town Airport
2 2 FlyFly Manchester North Airport Ho Chi Minh City Airport
3 3 AirEasy San Juan Island Airport New York Liberty Airport
4 4 FlyFly Rotterdam Harbor Airport Bogota Andes Airport
5 5 FlyHigh Frankfurt Main Airport Porto North Airport
6 6 FlyFly 200 rows Southern Star Airport
Database Consoles
postgres@localhost
console [postgres@localhost]
75:66
CRLF
UTF-8
4 spaces

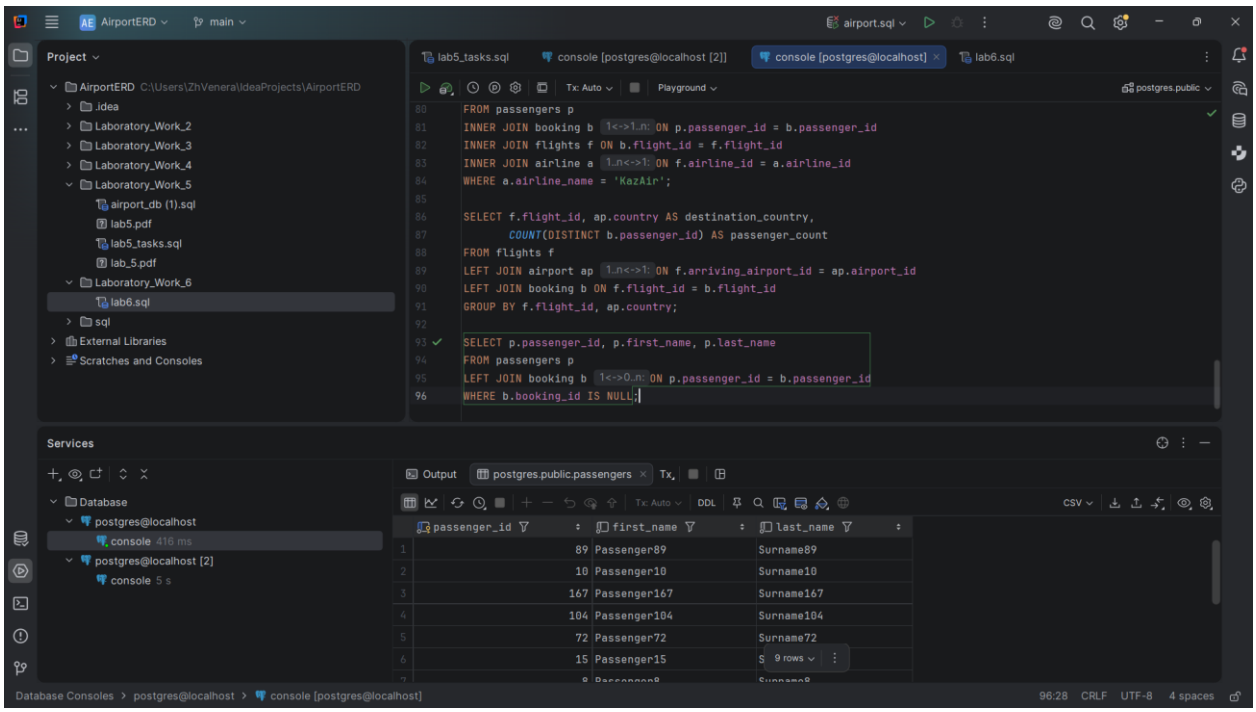
11)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
SELECT f.flight_id, a.airline_name,
dep.airport_name AS departure_airport,
arr.airport_name AS arrival_airport
FROM flights f
INNER JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
INNER JOIN airport dep 1..n<->1: ON f.departing_airport_id = dep.airport_id
INNER JOIN airport arr 1..n<->1: ON f.arriving_airport_id = arr.airport_id;
SELECT DISTINCT p.passenger_id,
CONCAT(p.first_name, ' ', p.last_name) AS full_name,
f.flight_id
FROM passengers p
INNER JOIN booking b 1<->1..n: ON p.passenger_id = b.passenger_id
INNER JOIN flights f ON b.flight_id = f.flight_id
INNER JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
WHERE a.airline_name = 'KazAir';
Services
Output
Result 14
Tx
CSV
passenger_id
full_name
flight_id
1 7 Passenger7 Surname7 7
2 17 Passenger17 Surname17 17
3 22 Passenger22 Surname22 22
4 23 Passenger23 Surname23 23
5 27 Passenger27 Surname27 27
6 32 Passenger32 Surname32 32
51 rows
Database
postgres@localhost
console 402 ms
postgres@localhost [2]
console 5 s
Database Consoles
postgres@localhost
console [postgres@localhost]
84:33
CRLF
UTF-8
4 spaces

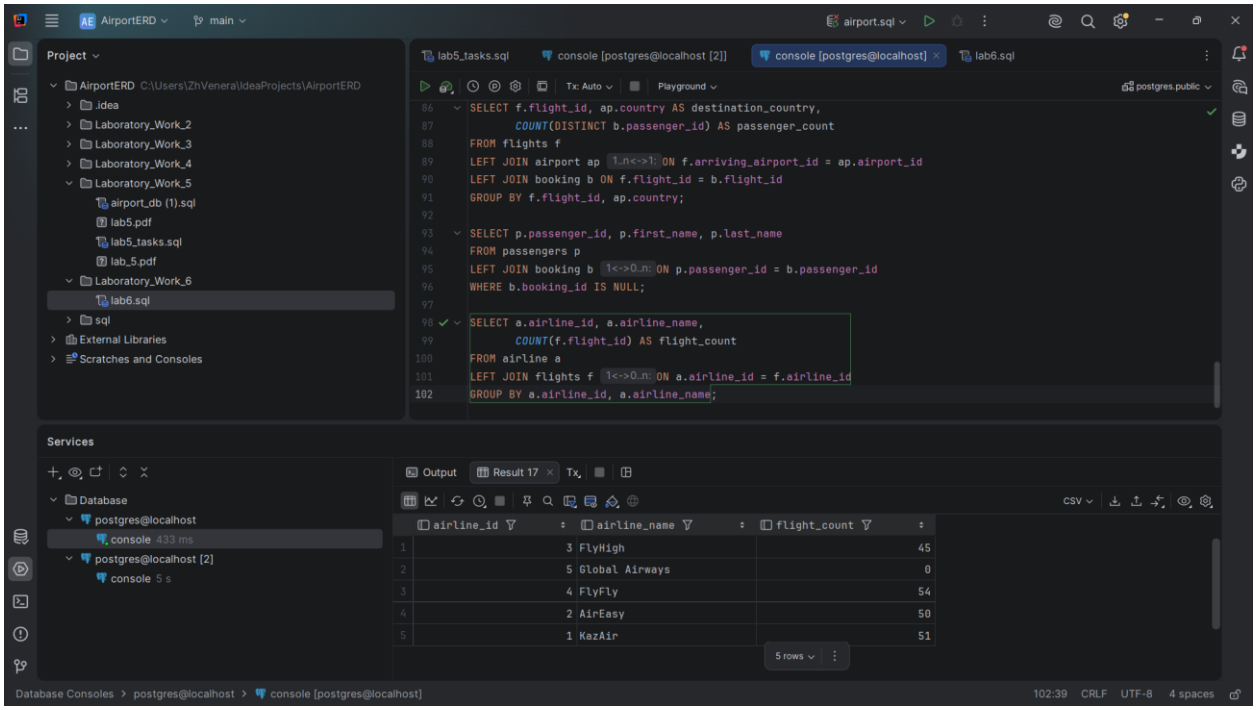
12)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
INNER JOIN airport arr 1..n<->1: ON f.arriving_airport_id = arr.airport_id;
SELECT DISTINCT p.passenger_id,
CONCAT(p.first_name, ' ', p.last_name) AS full_name,
f.flight_id
FROM passengers p
INNER JOIN booking b 1<->1..n: ON p.passenger_id = b.passenger_id
INNER JOIN flights f ON b.flight_id = f.flight_id
INNER JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
WHERE a.airline_name = 'KazAir';
SELECT f.flight_id, ap.country AS destination_country,
COUNT(DISTINCT b.passenger_id) AS passenger_count
FROM flights f
LEFT JOIN airport ap 1..n<->1: ON f.arriving_airport_id = ap.airport_id
LEFT JOIN booking b ON f.flight_id = b.flight_id
GROUP BY f.flight_id, ap.country;
Services
Output
Result 15
Tx
CSV
flight_id
destination_country
passenger_count
1 1 Lithuania 1
2 2 Vietnam 1
3 3 United States 1
4 4 Colombia 1
5 5 Portugal 1
6 6 Kazakhstan 1
200 rows
Database
postgres@localhost
console 430 ms
postgres@localhost [2]
console 5 s
Database Consoles
postgres@localhost
console [postgres@localhost]
91:34
CRLF
UTF-8
4 spaces

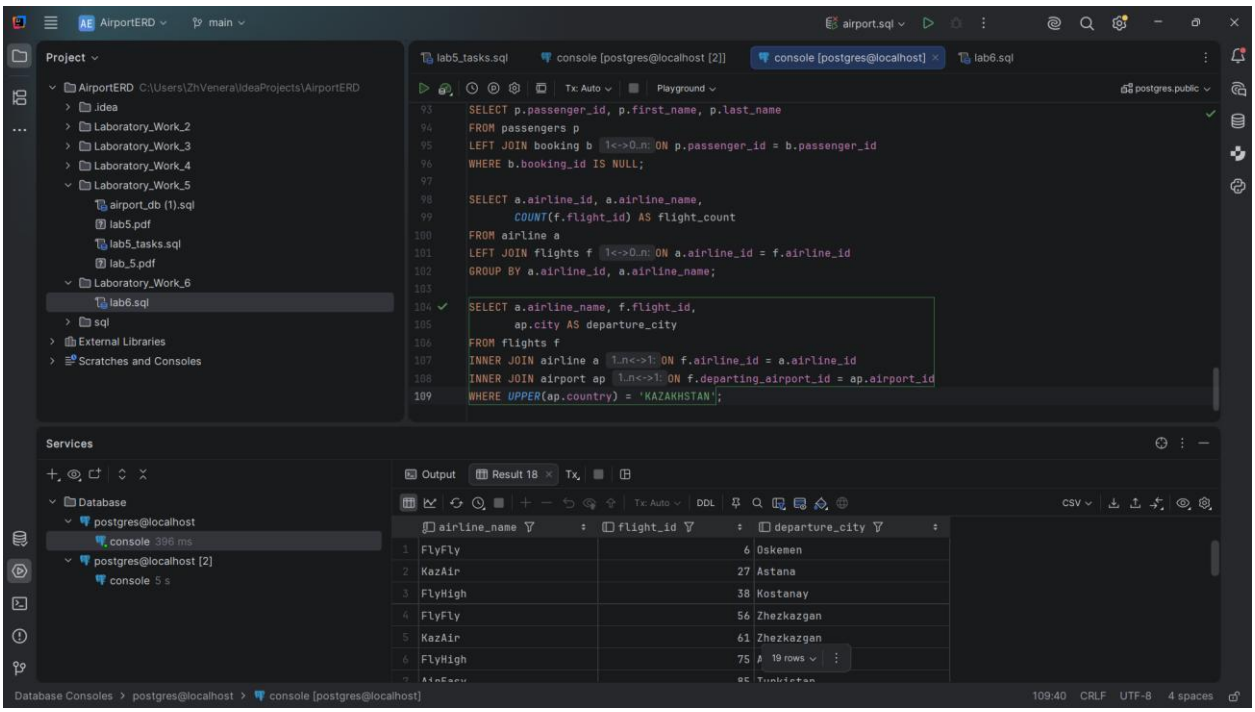
13)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
FROM passengers p
INNER JOIN booking b ON p.passenger_id = b.passenger_id
INNER JOIN flights f ON b.flight_id = f.flight_id
INNER JOIN airline a ON f.airline_id = a.airline_id
WHERE a.airline_name = 'KazAir';
SELECT f.flight_id, ap.country AS destination_country,
COUNT(DISTINCT b.passenger_id) AS passenger_count
FROM flights f
LEFT JOIN airport ap ON f.arriving_airport_id = ap.airport_id
LEFT JOIN booking b ON f.flight_id = b.flight_id
GROUP BY f.flight_id, ap.country;
SELECT p.passenger_id, p.first_name, p.last_name
FROM passengers p
LEFT JOIN booking b ON p.passenger_id = b.passenger_id
WHERE b.booking_id IS NULL;
Services
Output
postgres.public.passengers
passenger_id
first_name
last_name
89 Passenger89 Surname89
10 Passenger10 Surname10
167 Passenger167 Surname167
104 Passenger104 Surname104
72 Passenger72 Surname72
15 Passenger15 Surname15
9 rows
Database Consoles
postgres@localhost
console [postgres@localhost]
96:28
CRLF
UTF-8
4 spaces

14)


AirportERD
main
airport.sql
Project
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
SELECT f.flight_id, ap.country AS destination_country,
COUNT(DISTINCT b.passenger_id) AS passenger_count
FROM flights f
LEFT JOIN airport ap ON f.arriving_airport_id = ap.airport_id
LEFT JOIN booking b ON f.flight_id = b.flight_id
GROUP BY f.flight_id, ap.country;
SELECT p.passenger_id, p.first_name, p.last_name
FROM passengers p
LEFT JOIN booking b ON p.passenger_id = b.passenger_id
WHERE b.booking_id IS NULL;
SELECT a.airline_id, a.airline_name,
COUNT(f.flight_id) AS flight_count
FROM airline a
LEFT JOIN flights f ON a.airline_id = f.airline_id
GROUP BY a.airline_id, a.airline_name;
Services
Output
Result 17
airline_id
airline_name
flight_count
3 FlyHigh 45
5 Global Airways 0
4 FlyFly 54
2 AirEasy 50
1 KazAir 51
5 rows
Database Consoles
postgres@localhost
console [postgres@localhost]
102:39
CRLF
UTF-8
4 spaces

15)


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql
lab5.pdf
lab5_tasks.sql
lab_5.pdf
Laboratory_Work_6
lab6.sql
sql
External Libraries
Scratches and Consoles
lab5_tasks.sql
console [postgres@localhost [2]]
console [postgres@localhost]
lab6.sql
Tx: Auto
Playground
postgres.public
SELECT p.passenger_id, p.first_name, p.last_name
FROM passengers p
LEFT JOIN booking b 1<->0..n: ON p.passenger_id = b.passenger_id
WHERE b.booking_id IS NULL;
SELECT a.airline_id, a.airline_name,
COUNT(f.flight_id) AS flight_count
FROM airline a
LEFT JOIN flights f 1<->0..n: ON a.airline_id = f.airline_id
GROUP BY a.airline_id, a.airline_name;
SELECT a.airline_name, f.flight_id,
ap.city AS departure_city
FROM flights f
INNER JOIN airline a 1..n<->1: ON f.airline_id = a.airline_id
INNER JOIN airport ap 1..n<->1: ON f.departing_airport_id = ap.airport_id
WHERE UPPER(ap.country) = 'KAZAKHSTAN';
Services
Database
postgres@localhost
console 396 ms
postgres@localhost [2]
console 5 s
Output
Result 18
Tx: Auto
DDL
CSV
airline_name
flight_id
departure_city
1 FlyFly 6 Oskemen
2 KazAir 27 Astana
3 FlyHigh 38 Kostanay
4 FlyFly 56 Zhezkazgan
5 KazAir 61 Zhezkazgan
6 FlyHigh 75 Aktau
19 rows
Database Consoles > postgres@localhost > console [postgres@localhost]
109:40 CRLF UTF-8 4 spaces